

long as it would under an equivalent spending plan that started during a bull market cycle. This observation is often labeled by the term “sequence-of-returns” risk and is used by many in the insurance industry to explain the importance of downside protection during the so-called retirement risk zone—the years just before and just after your wealth accumulation peak illustrated earlier in Figure 1.1.

I would like to illustrate how a bull or bear market cycle can impact the sustainability of your portfolio by appealing to ideas from basic high school trigonometry, also known as the mathematics of sine and cosine waves. And although your skills might be rusty after these many years, the main story should be accessible to all.

First, allow me to review the basic arithmetic of generating income and sustainability. Assume you start retirement with a nest egg of exactly \$100 and you allocate this to an investment fund that earns a real (that is, after inflation) 5% per annum during every year of your retirement. For simplicity, I take this 5% to be an annual percentage rate (APR) that compounds continuously over time. Remember, this implies that if inflation is 3% per annum, then your nominal return is (approximately)  $5\% + 3\% = 8\%$ . If inflation is 4%, then your nominal return is (approximately) 9%.

Now let's spend some money. If you withdraw \$6 (also inflation adjusted) per year from this portfolio, the nest egg will be exhausted in exactly 35.8 years. In contrast, if you withdraw \$7 (inflation adjusted) per year, the funds will last for 25.1 years. If you withdraw \$8, the funds will last 19.6 years. Remember, all of this assumes your portfolio earns the same consistent inflation-adjusted 5% APR, for as long as the funds exist.

At this point I need you to suspend your disbelief and imagine a perfectly cyclical (sine wave) financial market—and I mean perfect with no randomness, noise, or real-world uncertainty. I'm going to describe two symmetrically opposed scenarios. Look at Figure 6.2 for a picture that's worth the next 200 words.

In this figure, the smooth middle line represents the path of the investment portfolio, assuming that it earns a constant 5% each and every year and that you are also withdrawing \$7 per year on a continuous basis. Let this line serve as our benchmark. In contrast, the solid